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Protosilylation

Stable and Recyclable Copper Nanoclusters with Exposed Active Sites for Broad-Scope Protosilylation in Open Air

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Abstract: Despite recent advances in cluster-based catalysis for organic synthesis, the substrate scope of reactions catalyzed by metal nanoclusters is typically not superior to previously established catalytic systems. Herein, we develop new atomically precise copper nanoclusters for protosilylation, with scope expanding to alkenes and simple enynes that were not suitable for prior synthetic methodologies with traditional copper complexes. The involvement of a second copper center in the metal kernel during the migratory insertion step is thought to be responsible for the expanded scope. In addition, the reaction is highly compatible with water and can be carried out in open air rather than under inert gas protection. Mechanistic studies suggest that the cluster-catalyzed protosilylation proceeds in the absence of silyl radicals. The current findings demonstrate the potential of using metal nanoclusters for practical and sustainable chemical synthesis.

Introduction

Because of the excellent tunability of ligand scaffolds and kernel structures, as well as the distinct photochemical characteristics of metal cores, atomically precise metal nanoclusters have shown promise as effective catalysts for the development of organic chemical processes.[1] In recent years, considerable efforts have been made toward constructing catalytically active copper nanoclusters in gram scales at low cost for sustainable synthetic applications[2] While several types of cluster-catalyzed coupling reactions can be achieved with increased efficiency,[3] developing robust copper nanocluster catalysts that can significantly broaden reaction scope when compared to established catalytic systems based on conventional copper complexes remains a major challenge.

On account of its flat structure and strong chelating effects, we anticipated that the incorporation of maleonitriledithiolate (mnt) in cluster synthesis would produce stable copper nanoclusters with exposed active sites suitable for practical catalytic applications.[4] The presence of Cu–Cu bonds in nanoclusters could have an important impact on the elementary steps during catalysis, potentially overcoming synthetic limitations in catalytic organic transformations. Since protosilylation reactions are crucial for organic synthesis[5] and can be facilitated by copper complexes,[6] investigating them with novel copper nanoclusters gave us an ideal opportunity to test the hypothesis. Even though copper-catalyzed protosilylation of alkynes and enynes with (dimethylphenylsilyl)boronic acid pinacol ester (pinB–$SiMe_2Ph$), a readily available and moisture-stable silylboronate, has been extensively studied through the use of various ligands (Schemes 1a and 1b),[7] there is still room for improvement in terms of catalyst turnover numbers (TONs) and reaction operability.

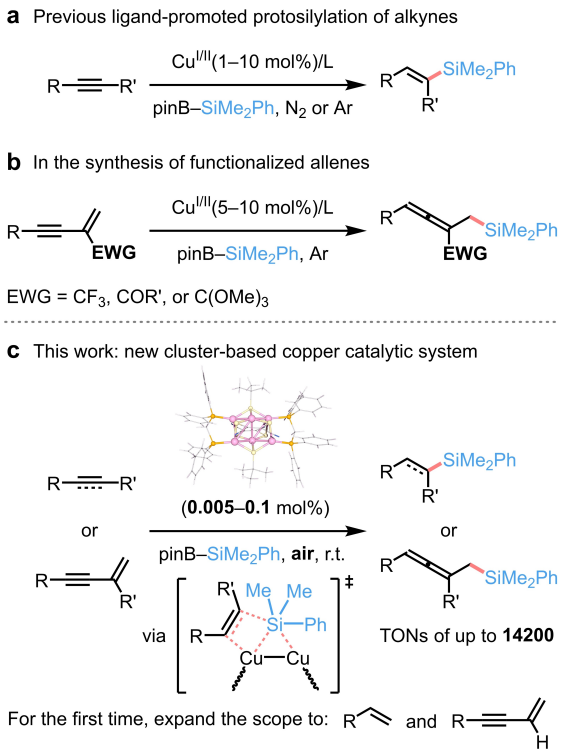
Here we report the synthesis and characterization of copper nanoclusters protected by mnt, thiolate, and bisphosphine ligands, which efficiently catalyze protosilylation of C–C multiple bonds under air atmosphere (Scheme 1c). The optimized nanocatalyst can be recycled multiple runs without loss of reactivity, and its remarkable stability is further demonstrated by UV–Vis, NMR, and ESI-MS studies. Aside from the high activity in gram-scale synthesis, the newly-developed copper nanocluster exhibits outstanding catalytic performance in the protosilylation of alkenes and enynes without the need for the pre-installation of electron-withdrawing functionalities.[7e,8] Additional interactions between the silyl group and the copper center adjacent to the exposed active site are likely to promote the key migratory insertion step, resulting in an expanded substrate scope in protosilylation reactions.

Results and Discussion

In a typical experiment, Cu_6 nanoclusters protected by mnt ligands were synthesized using a simple method that consists of two major steps: (i) synthesizing stable copper(I) precursor under ambient conditions and (ii) introducing additional protective ligands to induce the formation of

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Scheme 1. Copper-catalyzed protosilylation reactions.

monodisperse Cu_6 nanoclusters (*vide infra*). The Cu_6 nanoclusters were finally collected as orange-yellow crystalline materials by slow evaporation of solvents.

Step 1: Synthesis of the [Cu(^{t}BuS)]$_n$ precursor. A mixture of $[Cu(MeCN)_4]BF_4$ (0.10 mol, 32 g) and *tert*-butyl mercaptan (^{t}BuSH: 0.12 mol, 11.2 mL) were dissolved in acetonitrile (250 mL) under vigorous stirring at room temperature. Then, 5 mL of triethylamine was added to the reaction solution, and the reaction continued for 12 h at room temperature. Finally, yellow powder of [Cu(^{t}BuS)]$_n$ was collected by filtrating, washing with ethanol and ether, and drying under ambient environment (yield: 90 %).

Step 2: Synthesis of the Cu_6 nanoclusters. [Cu(^{t}BuS)]$_n$ (0.13 mmol, 20 mg) and bis(diphenylphosphino)methane (dppm) (0.039 mmol, 15 mg) were firstly added to a 6-mL mixed solution of acetonitrile and chloroform (1:2 v/v) under stirring. After 10 min, sodium maleonitriledithiolate (Na_2mnt: 0.06 mmol, 9 mg) and $NaBH_4$ (0.1 mmol, 4 mg) were added sequentially, and the reaction was allowed to proceed for 12 h, resulting in an organic-red solution. After 48 h, orange-yellow block crystals (**Cu_6-1**) can be obtained by solvent evaporation of filtrate in darkness with a yield of 61%. The synthesis of **Cu_6-2** was similar to that of **Cu_6-1**, with the exception of using 1,2-bis(diphenylphosphino)-ethane (dppe) instead of dppm.

The molecular composition and the charge state of the Cu_6 nanoclusters were confirmed by electrospray ionization mass spectrometry (ESI-MS) and single-crystal X-ray diffraction (SCXRD). As shown in Figure S1, ESI-MS revealed that **Cu_6-1** and **Cu_6-2** dissolved in chloroform had prominent peaks at mass/charge ratio (*m*/*z*) of 1630.7959 Da (calcd. 1630.7892 Da) and 1658.8438 Da (calcd. 1658.8206 Da), corresponding to $[\mathbf{Cu_6\text{-}1}+Na]^+$ and $[\mathbf{Cu_6\text{-}2}+Na]^+$ species, respectively. The isotopic patterns observed for both nanoclusters matched well with the simulated patterns. These results clearly showed that the whole structures of both **Cu_6-1** and **Cu_6-2** remained intact during the measurements, with charge states of 0. The components of the two nanoclusters are in line with the observed crystal structures (*vide infra*).

SCXRD analysis revealed that **Cu_6-1** crystallized in the monoclinic space group $P2_1$, while **Cu_6-2** crystallized in the monoclinic space group *Cc*, owing to the differences in the phosphine ligands utilized in the reaction systems. The formulas are determined as $[Cu_6(^t BuS)_2(dppm)_2(mnt)_2]$-•$CHCl_3$ and $[Cu_6(^t BuS)_2(dppe)_2(mnt)_2]$, respectively. Structural analysis of both nanoclusters, as presented in Figures 1a and 1f, indicates a comparable kernel structure and ligand surface. Specifically, **Cu_6-1** features a bowknot-like

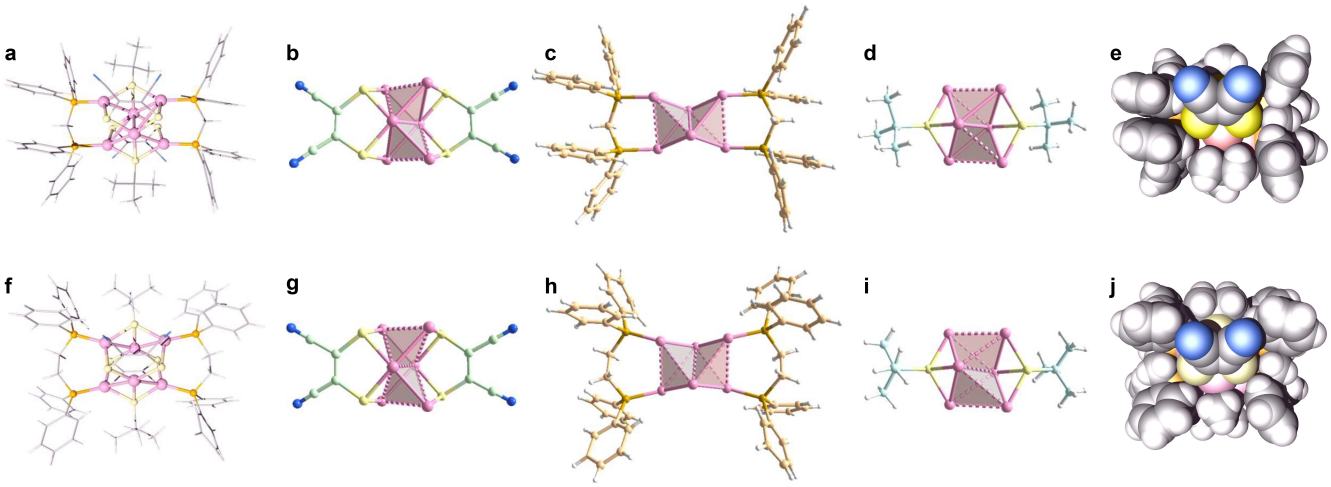


Figure 1. Crystal structures, ligand binding modes, and space-filling models of **Cu_6-1** (a–e) and **Cu_6-2** (f–j). Color labels: rose, Cu; gray, C; blue, N; tan, P; light yellow, S; white, H.

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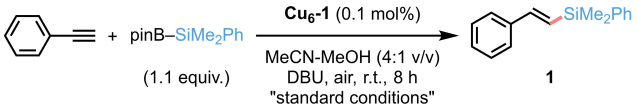
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Cu_6 kernel, which is capped by two mnt, dppm, and μ_3-^{t}BuS ligands with C_i symmetry (Figures 1b–1d). Each sulfur atom of the mnt ligand adopts a μ_2-binding mode, while both sulfur atoms of each mnt chelate with a potentially active copper center (Figures 1b). The remaining four copper atoms are bridged by the two bisphosphine ligands to form a stable nanocluster (Figures 1c). The Cu_6 kernel can be identified as a fusion of two Cu_4 tetrahedra that share one Cu–Cu edge. The Cu···Cu distances range from 2.663 to 2.983 Å, with an average of 2.799 Å. In comparison, **Cu_6-2** demonstrates a slightly different Cu_6 kernel. When the surface dppm is replaced by its homologue dppe, the shared Cu–Cu bond is stretched to break, as displayed in the kernel of **Cu_6-2** (Figures 1g–1i). Integrating the remaining Cu···Cu interactions reveals interatomic distances ranging from 2.613 to 2.967 Å, with an average value of 2.715 Å. This suggests that the length of the bisphosphine's side-chain is critical in determining the kernel shape of metal nanoclusters.

The space-filling models reveal that both of the kernels feature two exposed copper centers that are chelated by the mnt ligands (Figures 1e and 1j). The presence of the mnt ligands greatly enhances inter-cluster interactions. Strong $C(sp^2)$–H···π_{CN} interactions, along with other commonly observed bonding, were identified in **Cu_6-1** and **Cu_6-2**, according to the packing diagrams (Figures 2). These are likely to boost the cluster's stability for heterogeneous catalytic applications. It is noteworthy that the surface ligands of **Cu_6-1** exhibit shorter $C(sp^2)$–H···π_{CN} distances and stronger $C(sp^2)$–H···π_{Ph} interactions than **Cu_6-2**, indicating improved stability in catalysis.

Because of the exposed coordination sites and extra stability from inter-cluster interactions, we hypothesized that both copper nanoclusters (**Cu_6-1** and **Cu_6-2**) could act as potentially active catalysts for protosilylation reactions, which entail the formation of a new Cu–Si bond in the first elementary step.[7f,9] After testing a variety of reaction parameters, we discovered a robust synthetic procedure that allows for the cluster-catalyzed protosilylation of phenylacetylene with a TON of 910 (Table 1, entry 1; 91% yield).

Table 1: Copper-nanocluster-catalyzed protosilylation of phenylacetylene: effect of reaction parameters.[a,b]



Entry	Change from "standard conditions"	Yield [%]	TON[c]
1	none	91	910
2	no catalyst	<1	–
3	no base	<1	–
4	**Cu_6-2**, instead of **Cu_6-1**	85	850
5	**Cu_8**, instead of **Cu_6-1**	10	100
6	**Cu_{15}**, instead of **Cu_6-1**	13	130
7	**Cu_{54}**, instead of **Cu_6-1**	16	160
8[d]	$[Cu(MeCN)_4]BF_4$, instead of **Cu_6-1**	30	6.0
9[d]	CuCl, instead of **Cu_6-1**	30	6.0
10[d]	CuOAc, instead of **Cu_6-1**	29	5.8
11[d]	CuOTf, instead of **Cu_6-1**	30	6.0
12[d]	$Cu(OAc)_2$, instead of **Cu_6-1**	25	5.0
13[d]	$Cu(OTf)_2$, instead of **Cu_6-1**	30	6.0
14	NEt_3, instead of DBU	43	430
15	DMAP, instead of DBU	23	230
16	Na_2CO_3, instead of DBU	83	830
17	NaO^{t}Bu, instead of DBU	85	850
18	MeCN as solvent	29	290
19	MeOH as solvent	76	760
20	MeCN-H_2O (4:1 v/v) as solvent	73	730

[a] Standard conditions: phenylacetylene (0.30 mmol, 1.0 equiv.), $PhMe_2SiBpin$ (1.1 equiv.), **Cu_6-1** (0.1 mol%), and DBU (1.1 equiv.) in MeCN (2.4 mL) and MeOH (0.6 mL) under air atmosphere at room temperature for 8 h. [b] Yield was determined by ^{1}H NMR of the crude product using mesitylene as an internal standard. [c] TONs were calculated based on the crude NMR yield of 1. [d] Copper catalyst (5.0 mol%) was used. **Cu_8**, $[Cu_8H(9H\text{-carbazole-9-carbodithioate})_6](PF_6)$;[10] **$Cu_{15}$**, $[Cu_{15}(PET)_{13}(PPh_3)_6](BF_4)_2$;[11] **$Cu_{54}$**, $Cu_{54}S_{13}O_6(^t\text{BuS})_{20}(^t\text{BuSO}_3)_{12}$;[12] DMAP, 4-dimethylaminopyridine.

Control experiments confirmed that both the catalyst and the base are necessary for the reaction to proceed (entries 2 and 3). In comparison to **Cu_6-1**, **Cu_6-2** with less exposure to copper active sites displayed somewhat lower reactivity,

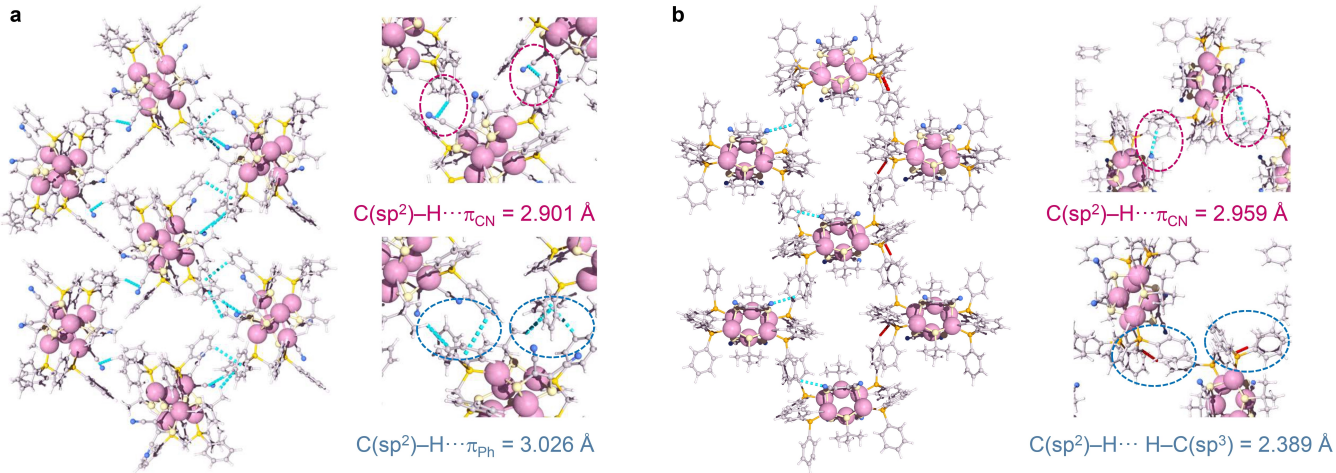


Figure 2. Packing structures of **Cu_6-1** (a) and **Cu_6-2** (b) in single crystals along the *bc* plane. Inset: inter-cluster interactions between the protective shells are highlighted as green or red dotted lines. Color labels: rose, Cu; gray, C; blue, N; tan, P; yellow, S; white, H.

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providing product **1** in 85% yield (entry 4). Nevertheless, these two newly developed nanocatalysts are significantly superior to other copper nanoclusters in terms of promoting protosilylation (entries 5–7). The conventional copper complexes, including various Cu(I) and Cu(II) salts as well as their synthetic precursor $[Cu(MeCN)_4]BF_4$, proved to be far less effective than the cluster-based catalysts (entries 8–13). While weak organic bases (e.g., NEt_3 and 4-dimethylaminopyridine) are not as good as 1,8-diazabicyclo[5.4.0]-undec-7-ene (DBU), inorganic bases, such as Na_2CO_3 and NaO^{t}Bu, also gave satisfactory results (entries 14 and 15). According to solvent studies, the involvement of protic solvents is crucial in the catalytic system (entries 18–20).

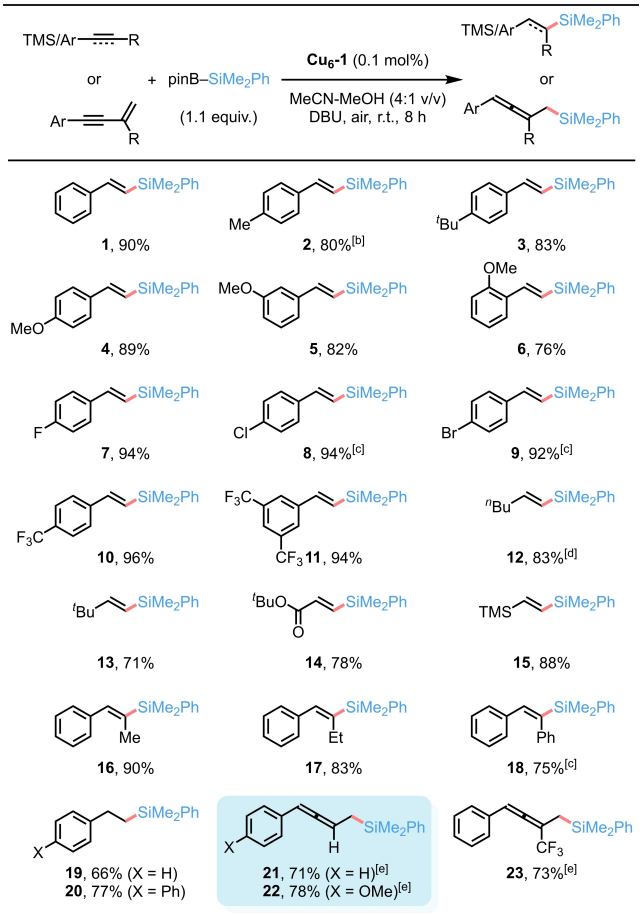
Under the optimized reaction conditions, the alkyne substrates containing a variety of substituted aryl rings furnish the desired (*E*)-products with excellent levels of regioselectivity (Table 2, products **1**–**11**). Both electron-donating and electron-withdrawing functional groups at different positions are well tolerated. Apart from phenylacetylene derivatives, alkyl-substituted terminal alkynes prove to be effective substrates in the protosilylation (Table 2, products **12** and **13**), despite the relatively low regioselectivity for the substrate with a primary alkyl chain. Furthermore, under standard conditions, an α,β-unsaturated ester (**14**) is obtained with high levels of regio- and stereoselectivity. Trimethylsilylacetylene, a commonly used organosilicon compound, could be converted into a *trans*-1,2-disilylethene (**15**) with 88% yield. In addition, internal alkyne substrates are highly compatible with the cluster-based catalytic system (Table 2, products **16**–**18**). Very importantly, the substrate scope can be further expanded to alkenes (Table 2, products **19** and **20**), which was difficult to achieve in previous copper catalysis (Figure S5). For the synthesis of silyl-functionalized allenes, we broadened the substrate scope from 2-trifluoromethyl enynes to simple ones (Figure S6), giving desired di-substituted allenes **21** and **22** in yields of 71% and 78%, respectively.

We have expanded our investigation on the functional-group compatibility of this synthetic method by conducting the template reaction in the presence of different additives (Table S7). Across a diverse array of additives such as aldehydes, ketones, nitriles, and various heterocycles, the cluster-catalyzed protosilylation proceeded with minimal or negligible influence on the yield of product **1**. While an aniline displayed some incompatibility, it remained stable under the reaction conditions. In contrast, although the protosilylation proceeded effectively in the presence of a phenol or an *N*-methyl pyrrole, these two additives exhibited poor recovery.

Notably, the gram-scale reaction can proceed smoothly with a 20-fold lower catalyst loading, providing a TON as high as 14200 (eq. 1). To demonstrate the synthetic utility of this protosilylation, we subjected product **1** to a range of coupling and hydrogenation conditions for the construction of carbon–carbon and carbon–heteroatom bonds (Scheme 2). A number of functionalized *trans*-olefins, including high value-added bromoalkene **26** and diene **28**, were synthesized in high yields. When an electron-deficient aryl group was introduced to the alkyne terminal of the enyne, a second protosilylation reaction occurred in the resulting di-substitute allene,[13] activating the double bond closer to the aromatic ring (eq. 2). The other silyl group approached the central carbon atom of the allene from the opposite direction as the previously installed silyl group, giving disilylated product **33** in the (*E*)-configuration, most likely through a migratory insertion pathway.

Prior to exploring the reaction mechanism, we systematically evaluated the recyclability of the nanocluster catalysts. Impressively, the optimal catalyst (**Cu_6-1**) retained its reactivity for protosilylation after 8 runs, whereas **Cu_6-2** with relatively weaker inter-cluster interactions tended to lose the catalytic activity in the 8th run (Figure 3a). Similar trends were observed in other heterogeneous catalysis involving inorganic bases (Tables S5 and S6). The UV–Vis absorption spectra, along with the ^{1}H NMR, ^{31}P NMR, and ESI-MS data, confirmed that **Cu_6-1** had the same chemical composition before and after catalysis (Figure 3b and Figure S7–S9),

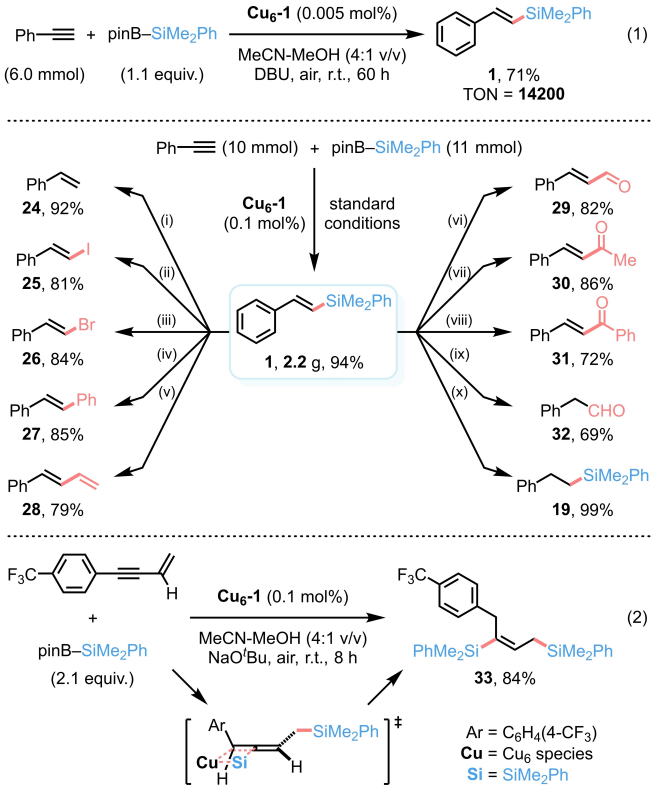
Table 2: Substrate scope for protosilylation with **Cu_6-1**.[a]



[a] Reaction conditions: alkyne, alkene, or enyne (0.30 mmol, 1.0 equiv.), $PhMe_2SiBpin$ (1.1 equiv.), **Cu_6-1** (0.1 mol%), and DBU (1.1 equiv.) in MeCN (2.4 mL) and MeOH (0.6 mL) under air atmosphere at room temperature for 8 h. For each entry number (in bold), data are reported as isolated yields. Without additional notice, the regioisomeric ratios (r.r.) were found to exceed 20:1, as determined by ^{1}H NMR analysis of the crude reaction mixture. [b] **Cu_6-1** (0.2 mol%) was used. [c] 12 h. [d] r.r. = 2.2:1. [e] NaO^{t}Bu (1.1 equiv.) as base.

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Scheme 2. Gram-scale synthesis and further synthetic applications. Reaction conditions: (i) HCl (conc.), MeCN, 60 °C, 24 h; (ii) NIS, MeCN, r.t., 12 h; (iii) NBS, MeCN, r.t., 12 h; (iv) cat. $Pd(dba)_2$, iodobenzene, TBAF, THF, r.t., 2 h; (v) cat. $Pd(OAc)_2$, bromoethene, THF, 150 °C, 24 h; (vi) dichloromethyl methyl ether, $TiCl_4$, DCM, –78 °C, 6 h; (vii) acetyl chloride, $AlCl_3$, DCE, r.t., 3 h; (viii) benzoyl chloride, $TiCl_4$, DCM, –78 °C, 1 h; (ix) cat. $Pd(OAc)_2$, iodobenzene diacetate, AcOH, 120 °C, 24 h; (x) cat. Pd/C, H_2, EA, r.t., 12 h.

suggesting neither catalyst decomposition nor ligand exchange occurred during the catalytic protosilylation reaction. Due to the high compatibility with air atmosphere and the lack of influence from (2,2,6,6-tetramethylpiperidin-1-yl)oxy (TEMPO), a well-established radical trapping reagent (Figure 3c), we concluded that the protosilylation pathway involving a silyl radical intermediate[14] is highly unlikely, which clearly differs from cluster-based catalysis for protoborylation.[15] Since a mixed-solvent strategy was employed in the reaction optimization, we examined the proton source in the catalytic system with the corresponding deuterated solvents (Figure 3d). When acetonitrile-d_3 was utilized instead of acetonitrile, no deuterated product formed. The deuterium ratio of D_a near the aromatic substituent increased to 67% and 77%, when the reaction was carried out under air and nitrogen atmospheres, respectively. The unexpected deuteration of D_b, albeit to a limited extent, implied that a cluster–alkynyl species or an acetylide anion may be produced and then protonated under the reaction conditions (Table S8). The greater deuterium ratios of both D_a and D_b under nitrogen atmosphere suggested that water in the reaction mixture could possibly contribute to the protonation steps.

Based on the experimental results and DFT calculations, we proposed a plausible catalytic cycle in Figure 3e. With the aid of the base and methanol, the exposed active site on the copper nanocluster undergoes ligand association to generate cluster–silyl intermediate **A**. Following the side-on binding of an alkyne molecule through transition state **TS-1**, intermediate **B** forms with a ^{t}BuS ligand dissociated from the copper active center. In the subsequent silyl migration step, 1,2-insertion occurs with an additional stabilization effect from the nearby copper center, resulting in the highest activation energy of only 23.1 kcal mol^{-1}. Because of this unique migratory insertion mode involving two copper centers of **TS-2**, the catalytic protosilylation proceeds smoothly at room temperature, with an extended substrate scope that includes C=C double bonds. Furthermore, the concerted insertion leads to the formation of intermediate **C**, in which the cluster unit and the silyl group are *cis* to each other. Therefore, deprotonation of **C** with methanol yields a *trans*-alkene and intermediate **D**. The initial cluster catalyst (**Cu_6-1**) is regenerated via ligand dissociation in the presence of a conjugate acid.

Conclusion

We present here the development of mnt ligand-protected Cu_6 nanoclusters with exposed active sites for efficiently catalyzing protosilylation reactions under ambient conditions, with a TON of up to 14200. The nanocatalyst can be recycled at least 8 runs and is applicable to gram-scale synthesis. Apart from its high reactivity with alkyne substrates, the Cu_6 nanocluster can also be employed to functionalize alkenes and simple enynes. The new migratory insertion mode based on two copper centers in the cluster kernel provides guidance for the future design of catalytically active metal nanoclusters to transform synthetic methodologies.

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Conflict of Interest

The authors declare no conflict of interest.

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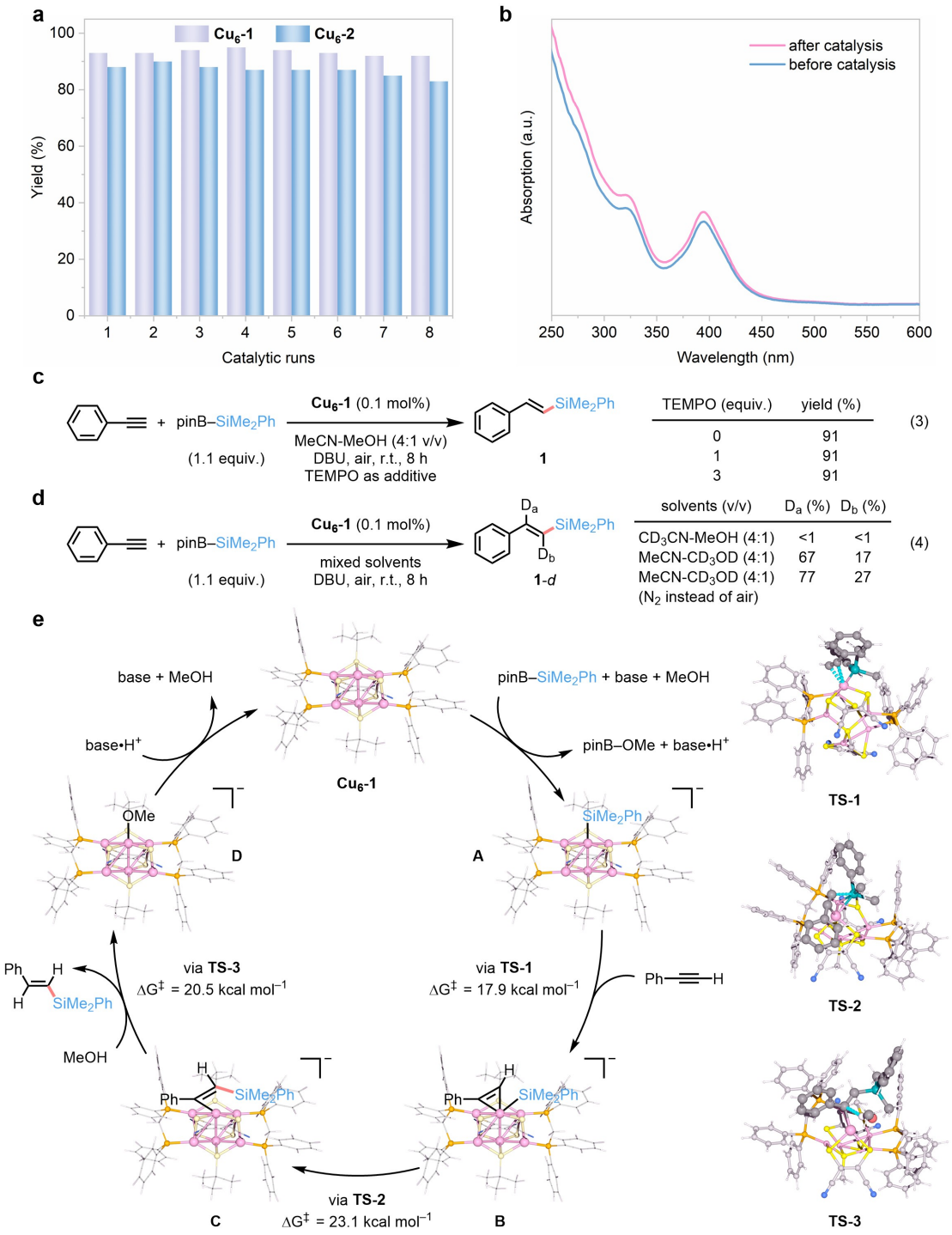


Figure 3. (a) Recycling experiments for cluster-catalyzed protosilylation of phenylacetylene. (b) UV – Vis absorption spectra of **Cu_6-1** before and after catalysis. (c) Influence of TEMPO. (d) Deuterium labelling experiments. (e) Plausible mechanism for protosilylation with **Cu_6-1**.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords: Copper · Ligand Design · Metal Nanoclusters · Protosilylation · Sustainable Synthesis

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[16] CCDC 2323892 and 2323893 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data request/cif, or by emailing data request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

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